

**DEPARTMENT OF MATHEMATICS & STATISTICS**

**MTH 211: Theory of Statistics**

(Session: 2025-26      Semester: II)

**Problem Set 5**

(Testing of Hypotheses – Preliminaries, MP test)

1. Let  $X$  be a random observation from

$$f_{\theta}(x) = \begin{cases} \theta x^{\theta-1}, & \text{if } 0 < x < 1 \\ 0, & \text{otherwise,} \end{cases}$$

$\theta \in \{1, 2\}$ . For testing  $H_0: \theta = 1$  against  $H_A: \theta = 2$ , based on a sample of size 2, the critical region is  $\omega := \{(x_1, x_2): \frac{1}{2x_1} \leq x_2\}$ . Find the size and power of the test.

2. Let  $X_1$  and  $X_2$  be a random sample from

$$f_{\theta}(x) = \begin{cases} \frac{1}{\theta} e^{-x/\theta}, & \text{if } x > 0 \\ 0, & \text{otherwise,} \end{cases}$$

$\theta \in \{1, 2\}$ . If the critical region for testing  $H_0: \theta = 2$  against  $H_A: \theta = 1$  is

$$\omega := \left\{ (x_1, x_2): \frac{\prod_{i=1}^2 f_{\theta_0}(x_i)}{\prod_{i=1}^2 f_{\theta_1}(x_i)} \leq \frac{1}{2} \right\}. \text{ Find the size and power of the test.}$$

3. The standardized lifetime of an equipment is normally distributed with mean  $\theta$  and standard distribution 5. For testing the null hypothesis  $H_0: \theta \leq 30$  against the alternative  $H_1: \theta > 30$ , a random sample of size  $n$  is chosen. Determine the sample size  $n$  and cutoff value  $c$ , such that the test  $\varphi(x) = \begin{cases} 1, & \text{if } \bar{x} > c \\ 0, & \text{if } \bar{x} < c \end{cases}$  has power function values equal to 0.1 and 0.9 at  $\theta = 30$  and  $\theta = 35$ , respectively. (Given  $\Phi(1.282) = 0.9$ )

4. Based on a random sample  $X_1, \dots, X_n$ , derive the MP size- $\alpha$  test for testing  $H_0: \theta = \theta_0$  against  $H_A: \theta = \theta_1 (> \theta_0)$  for the following populations:

(a)  $f_{\theta}(x) = \begin{cases} \frac{1}{\theta} e^{-x/\theta}, & \text{if } x > 0, \theta > 0 \\ 0, & \text{otherwise.} \end{cases}$

(b)  $f_{\theta}(x) = \begin{cases} \frac{e^{-\theta} \theta^x}{x!}, & \text{if } x = 0, 1, 2, \dots, \theta > 0 \\ 0, & \text{otherwise.} \end{cases}$

(c)  $f_{\theta}(x) = \begin{cases} \frac{\theta}{x^{\theta+1}}, & \text{if } x \geq 1, \theta > 0 \\ 0, & \text{otherwise.} \end{cases}$

(d)  $f_{\theta}(x) = (\sqrt{2\pi}\theta)^{-1} e^{-x^2/2\theta^2}, -\infty < x < \infty, \theta > 0$

(e)  $f_{\theta}(x) = (\sqrt{2\pi})^{-1} e^{-(x-\theta)^2/2}, -\infty < x < \infty, -\infty < \theta < \infty$

5. Let  $X$  be an observation in  $(0,1)$ . Find an MP size- $\alpha$  test for testing  $H_0: X \sim f_0$  against  $H_A: X \sim f_1$ , where,

$$f_0(x) = \begin{cases} 4x & 0 < x < \frac{1}{2} \\ 4-4x & \frac{1}{2} \leq x < 1 \end{cases} \quad \text{and} \quad f_1(x) = 1; 0 < x < 1.$$

6. Let  $X_1$  and  $X_2$  be i.i.d. observations with a pdf  $f(x)$ , which can be either of the following two pdfs, viz.,

$$f_0(x) = \begin{cases} 3x^2/64, & \text{if } 0 < x < 4 \\ 0, & \text{otherwise,} \end{cases} \quad \text{and} \quad f_1(x) = \begin{cases} 3\sqrt{x}/16, & \text{if } 0 < x < 4 \\ 0, & \text{otherwise.} \end{cases}$$

Show that the  $MP(\alpha)$  test for testing  $H_0: f(x) = f_0(x)$  against  $H_A: f(x) = f_1(x)$

rejects  $H_0$  if  $\log(2^{24}/(X_1 X_2)^6) > \chi_{4,\alpha}^2$ .

**Note:**  $Prob(Y > \chi_{4,\alpha}^2) = \alpha$ , where  $Y \sim \chi^2$  with 4 degrees of freedom.

7. Derive the MP size- $\alpha$  test for testing  $H_0: \theta = \theta_0$  against  $H_A: \theta = \theta_1 (< \theta_0)$  based on a random sample  $X_1, \dots, X_n$  from

$$f_{\theta}(x) = \begin{cases} \theta x^{\theta-1}, & \text{if } 0 < x < 1, \theta > 0 \\ 0, & \text{otherwise.} \end{cases}$$

8. Let  $X$  be a random variable with the following pmf's under  $H_0$  and  $H_A$ . Find the MP test at level  $\alpha = 0.04$ . Calculate the probability of Type II error of this test.

| x            | 1    | 2    | 3    | 4    | 5    | 6    | 7    |
|--------------|------|------|------|------|------|------|------|
| $f_{H_0}(x)$ | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.94 |
| $f_{H_1}(x)$ | 0.06 | 0.05 | 0.04 | 0.03 | 0.02 | 0.01 | 0.79 |